

3.5 Carbonyl Compounds

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.5 Carbonyl Compounds
Difficulty	Medium

Time allowed: 20

Score: /10

Percentage: /100

Question 1

$\text{CH}_3\text{COCH}_2\text{CHO}$ $\text{CH}_3\text{COCH}_2\text{CH}_2\text{OH}$ $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$

1 2 3 4

Which of these compounds gives a positive response to Tollens' reagent and can also be oxidised by acidified dichromate (VI) solution?

- A. 1 and 2
- B. 1 and 3
- C. 2 and 3
- D. 3 and 4

[1 mark]

Question 2

When alkaline I_2 solution is added to butanone, a yellow precipitate is formed.

Which of the following compounds is responsible for the precipitate?

- A. $\text{CH}_3\text{CH}_2\text{COCH}_2\text{I}$
- B. $\text{CH}_3\text{CH}_2\text{CO}_2^-$
- C. $\text{CH}_3\text{CHICOCH}_3$
- D. CHI_3

[1 mark]

Question 3

Liquid **L** is known to be either a single organic compound or a mixture of organic compounds.

Two tests were done on liquid **L** to determine its constituents:

- When treated with sodium, **L** gives off hydrogen gas.
- When treated with 2,4-dinitrophenylhydrazine reagent, **L** gives orange crystals.

Which deductions about **L** can definitely be made?

- 1** At least one component of **L** is an alcohol.
- 2** At least one component of **L** is a carbonyl compound.
- 3** Only one of the components of **L** is a carbonyl compound.

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 2 only

[1 mark]

Question 4

The colour of warm acidified sodium dichromate (VI) changes from orange to green when added to compound **Y**. 1 mol of **Y** reacts with 2 mol of hydrogen cyanide in the presence of potassium cyanide.

What could **Y** be?

A. $\text{H}_2\text{C}=\text{CHCH}_2\text{CHO}$

B. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$

C. $\text{OHCCH}_2\text{CH}_2\text{CHO}$

D. $\text{CH}_3\text{COCH}_2\text{COCH}_3$

[1 mark]

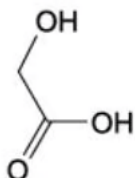
Question 5

Hydroxyethanal has the structural formula HOCH_2CHO .

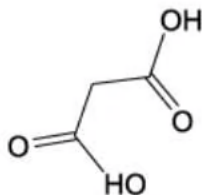
In an experiment hydroxyethanal is heated under reflux with an excess of acidified potassium dichromate (VI) until no further oxidation takes place.

What is the skeletal formula of the organic product formed in the experiment?

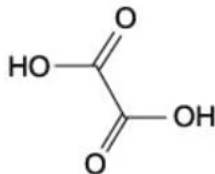
A.



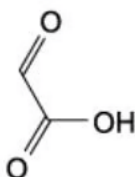
B.



C.



D.



[1 mark]

Question 6

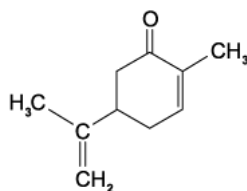
Which reagent will give similar results with both propanone and propanal?

- A. 2,4–dinitrophenylhydrazine reagent
- B. Acidified aqueous potassium dichromate (VI)
- C. An aqueous solution containing $[\text{Ag}(\text{NH}_3)_2]^+$ (Tollens' reagent)
- D. An alkaline solution containing complexed Cu^{2+} ions (Fehling's solution)

[1 mark]

Question 7

Carvone is used as a flavouring for spearmint chewing gum.



Carvone

When carvone is heated for a prolonged time with hot concentrated acidified potassium manganate (VII) the products carbon dioxide and compound **X** are formed.

X contains nine carbon atoms and reacts with 2,4–dinitrophenylhydrazine reagent.

What is the maximum number of molecules of 2,4–dinitrophenylhydrazine that will react with one molecule of **X**?

- A. 4
- B. 3
- C. 2
- D. 1

[1 mark]

Question 8

When ethanal and propanone are reacted separately with reagent X both cause the same colour change.

Which colour changes occurs in both reactions?

- 1 Black to yellow
- 2 Orange to green
- 3 Blue to red

- A. 1 only
- B. 1 and 2
- C. 2 and 3
- D. 1, 2 and 3

[1 mark]

Question 9

A reaction occurs between ethanal and hydrogen cyanide in the presence of sodium cyanide.

Which statements about the reaction mechanism are correct?

- 1 The last stage involves the formation of a hydrogen-oxygen bond.
- 2 In the intermediate, the oxygen carries a negative charge.
- 3 A new carbon-carbon bond is formed.

- A. 1 only
- B. 1 and 2
- C. 2 and 3
- D. 1, 2 and 3

[1 mark]

Question 10

When compounds 1, 2 and 3 react with PCl_5 , hydrogen chloride is evolved. However, only one of the three reacts with 2,4-dinitrophenylhydrazine reagent.

Which one of the following combinations could be 1, 2 and 3?

	1	2	3
A	$\begin{array}{c} \text{CH}_2\text{OH} \\ \\ \text{CH}_2\text{OH} \end{array}$	$\begin{array}{c} \text{CH}_2\text{OH} \\ \\ \text{CHO} \end{array}$	$\begin{array}{c} \text{CHO} \\ \\ \text{CO}_2\text{H} \end{array}$
B	$\begin{array}{c} \text{CH}_2\text{OH} \\ \\ \text{CHO} \end{array}$	$\begin{array}{c} \text{CH}_2\text{OH} \\ \\ \text{CO}_2\text{H} \end{array}$	$\begin{array}{c} \text{CHO} \\ \\ \text{CO}_2\text{H} \end{array}$
C	$\begin{array}{c} \text{CH}_2\text{OH} \\ \\ \text{CO}_2\text{H} \end{array}$	$\begin{array}{c} \text{CHO} \\ \\ \text{CO}_2\text{H} \end{array}$	$\begin{array}{c} \text{CO}_2\text{H} \\ \\ \text{CO}_2\text{H} \end{array}$
D	$\begin{array}{c} \text{CH}_2\text{OH} \\ \\ \text{CH}_2\text{OH} \end{array}$	$\begin{array}{c} \text{CHO} \\ \\ \text{CHO} \end{array}$	$\begin{array}{c} \text{CO}_2\text{H} \\ \\ \text{CO}_2\text{H} \end{array}$

[1 mark]

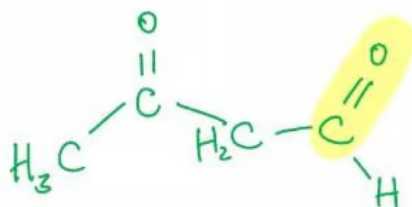
Model Answers: Medium

1

The correct answer is **B** because:

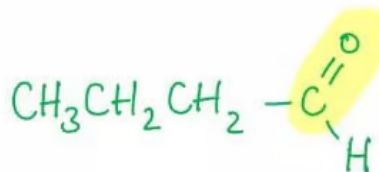
- Tollens' reagent is an aqueous solution of **silver nitrate** in **excess ammonia solution** used to test for the presence of **aldehyde** compounds.
- **Acidified potassium dichromate (VI)** is an **orange** oxidising agent which can oxidise **primary alcohols**, **secondary alcohols** and **aldehydes** to carboxylic acids, ketones and carboxylic acids respectively.
- Both **compounds 1** and **3** contain an aldehyde group which can react with both the Tollens' reagent and acidified potassium dichromate (VI).

Compound 1:

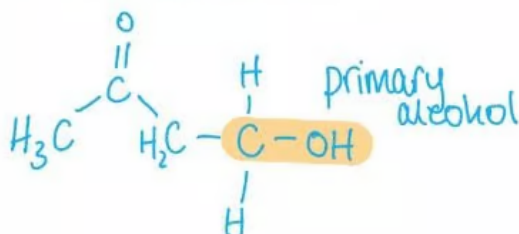


aldehyde groups react with Tollens' and KMnO_4

Compound 3:

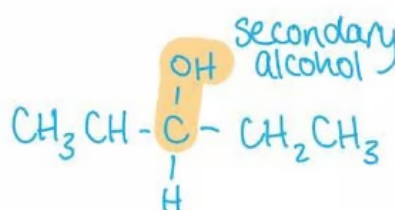


Compound 2:



alcohol groups only react with KMnO_4

Compound 4:



A & C are incorrect as **compound 2** contains a **ketone** and **primary alcohol** group. Only the alcohol is oxidised by potassium dichromate (VI) to a carboxylic acid under reflux,

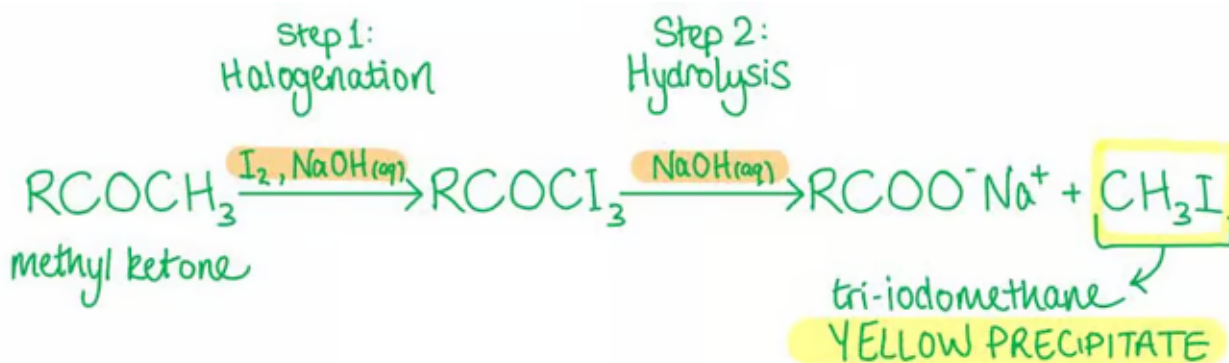
however the alcohol group will **not** react with Tollens' reagent.

D is incorrect as **compound 4** only contains a **secondary alcohol** which will be oxidised by potassium dichromate (VI) to a ketone but will not react with Tollens' reagent.

2

The correct answer is **D** because:

- **Trio-iodomethane** (also known as **iodoform**) forms a **yellow precipitate** with compounds containing a methyl ketone, **CH₃CO-**, group.
 - An alkaline solution of iodine is warmed with the substance to be tested.
 - The carbonyl compound is halogenated and the three hydrogen atoms in the CH₃ methyl group are replaced by iodine atoms.
 - This is a nucleophilic substitution reaction as the **iodine** acts as a **nucleophile** replacing the hydrogen atoms in CH₃.
 - An intermediate is **hydrolysed** to form the yellow precipitate of **tri-iodomethane**.
- The **black** iodine solution will contain **yellow crystals** of trio-iodomethane.



A is incorrect as the ketone has not fully halogenated.

B is incorrect as this is the salt formed from the hydrolysis of the **halogenated** ketone.

C is incorrect as halogenation takes place on the **methyl ketone**.

3

The correct answer is **D** because:

- The presence of a **carbonyl group** in an **aldehyde** or **ketone** can be tested by adding a solution of 2,4–dinitrophenylhydrazine (**2,4–DNPH**).
- If an aldehyde or ketone is present, a **deep–orange precipitate** is formed.
- The precipitate can be purified by **recrystallisation** and its **melting point** can be measured.
- The identity of the precipitated compound can then be found by referring to the **melting point data** from which the specific aldehyde or ketone can be identified.
- This reaction is an example of a **condensation reaction** in which two molecules join together and eliminate a small molecule (H_2O).
- Since reacting **L** with **2,4–DNPH** gave orange crystals, **L** could contain **either** an **aldehyde** or a **ketone**.
- Both **carboxylic acids** and **alcohols** (such as ethanol) can react with sodium (a reactive metal) to form a **salt** and **hydrogen gas**.
- Therefore, it is not certain that at least one of the components in **L** will be an **alcohol** as it could also be a **carboxylic acid**.

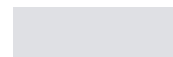
A, B & C are incorrect as a carboxylic acid can react with sodium to give a salt and **hydrogen gas** as well.

4

The correct answer is **C** because:

- **Acidified potassium dichromate (VI)** is an **orange** oxidising agent.
 - An **oxidising agent** is a species that **oxidises** another species and gets **reduced** itself.
 - The reaction mixture turns **green** as the orange dichromate ions ($\text{Cr}_2\text{O}_7^{2-}$) are reduced to **green** Cr^{3+} ions.
 - Only **primary alcohols**, **secondary alcohols** and **aldehydes** can be oxidised by acidified potassium dichromate (VI).
- **Compound Y** can therefore be either a **primary alcohol**, a **secondary alcohol** or an **aldehyde** as all these compounds will reduce the $\text{Cr}_2\text{O}_7^{2-}$ ions to Cr^{3+} ions.

- A **nucleophilic addition** is when a **nucleophile** is added to an unsaturated bond to form a **single** product.
 - A nucleophile is a species that donates a pair of electrons.
- **Hydroxynitriles** can be formed from the **nucleophilic addition** reaction of **aldehydes** and **ketones** with HCN, KCN as catalyst and heat.
- The cyanide ion, CN^- , acts as a **nucleophile** and is added on to the carbonyl carbon to form a single product only.



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- The cyanide ion, CN^- , acts as a **nucleophile** and is added on to the carbonyl carbon to form a single product only.
- Compound Y therefore contains an **aldehyde group**.
- Since **two moles of CN^-** react, compound Y must have **two aldehyde groups**.

A & B are incorrect as these compounds only contain **one aldehyde** group.

D is incorrect as this compound contains **two ketone** groups instead of two aldehyde groups. Though the ketone will react with hydrogen cyanide, it will **not** undergo oxidation by acidified potassium dichromate (VI).

5

The correct answer is **C** because:

- **Acidified potassium dichromate (VI)** is an **orange** oxidising agent.
 - An **oxidising agent** is a species that **oxidises** another species and gets **reduced** itself.
 - The reaction mixture turns green as the **orange $\text{Cr}_2\text{O}_7^{2-}$** ions (dichromate) are reduced to **green Cr^{3+}** ions.
 - Only **primary alcohols**, **secondary alcohols** and **aldehydes** can be oxidised by acidified potassium dichromate (VI).
- **Hydroxyethanal** contains a **primary alcohol** and **aldehyde** group.
- Since the compound is heated **under reflux** until no further oxidation takes place, both the primary alcohol and the aldehyde are oxidised to **carboxylic acid groups**.
 - Primary alcohol → aldehyde → carboxylic acid

o Aldehyde → carboxylic acid

- The correct answer is therefore **C**.

A is incorrect as **only** the **aldehyde** group has been oxidised.

B is incorrect as this compound contains the **incorrect** number of **carbon atoms** in the carbon chain (there should be only 2 carbon atoms and not 3).

D is incorrect as the molecule is not fully oxidised; the aldehyde can undergo further oxidation to a carboxylic acid.

6

The correct answer is **A** because:

- The presence of a **carbonyl group** in an **aldehyde** or **ketone** can be tested by adding a solution of 2,4-dinitrophenylhydrazine (**2,4-DNPH**).
- If an aldehyde or ketone is present, a **deep-orange precipitate** is formed.
- The precipitate can be purified by **recrystallisation** and its **melting point** can be measured.
- The identity of the precipitated compound can then be found by referring to the melting point data from which the specific aldehyde or ketone can be identified.
- This reaction is an example of a **condensation reaction** in which two molecules join together and eliminate a small molecule (H_2O).
- Therefore, 2,4-dinitrophenylhydrazine will form a **deep-orange precipitate** with both **propanone** (a ketone) and **propanal** (an aldehyde).

B is incorrect as **acidified potassium dichromate (VI)** is an **orange** oxidising agent which turns **green** as the **orange $\text{Cr}_2\text{O}_7^{2-}$ ions** (dichromate) are reduced to **green Cr^{3+} ions** when **primary alcohols**, **secondary alcohols** and **aldehydes** are oxidised by acidified potassium dichromate (VI). Therefore, **only propanal** will turn the solution **green** as ketones cannot be further oxidised.

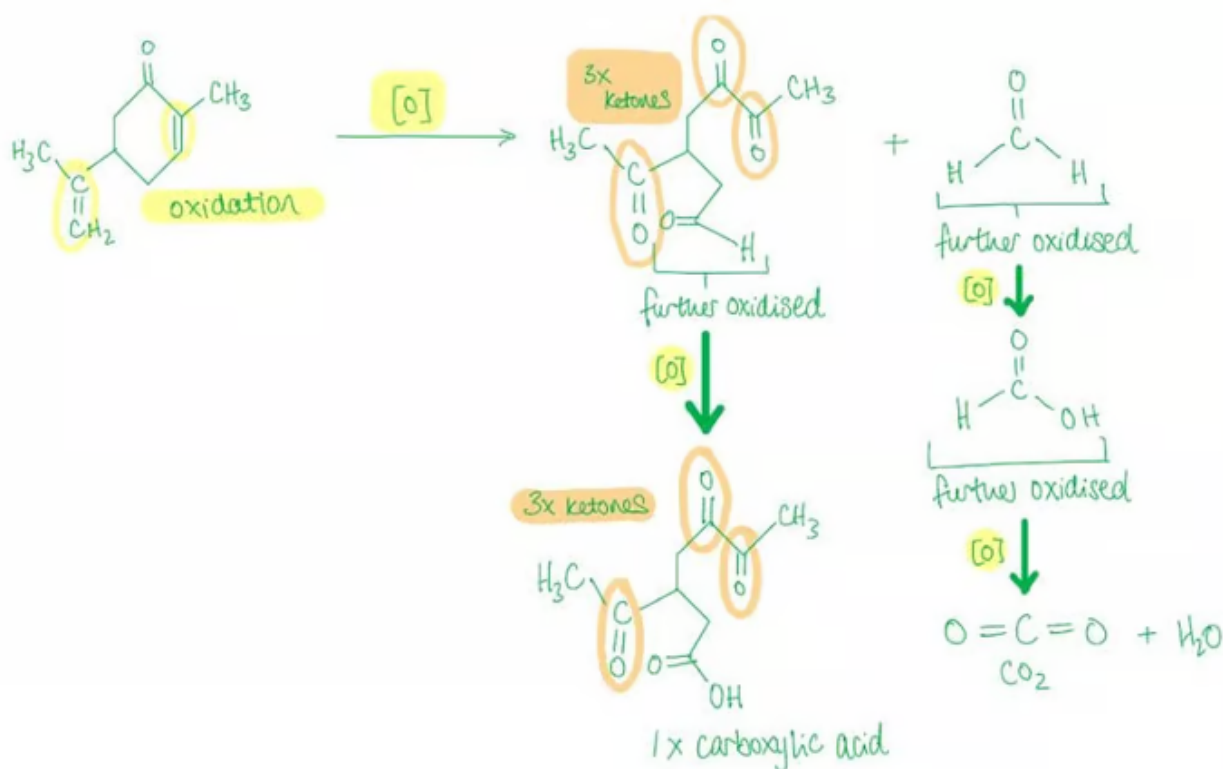
C is incorrect as Tollens' reagent is an aqueous solution of **silver nitrate** in **excess ammonia solution** used to test for the presence of **aldehyde** compounds. Therefore, **only propanal** will result in the formation of a '**silver mirror**' on the inside of the tube.

D is incorrect as the Fehling's solution is an **alkaline** solution containing copper (II) ions which tests for the presence of aldehydes. Therefore, **only propanal** will change the clear **blue** colour of the Fehling's solution **to red**.

7

The correct answer is **B** because:

- **Oxidation** in organic chemistry is the **gain of oxygen** atoms.
- The acidified potassium manganate (VII) solution oxidises the **alkene** by **breaking** the C=C double bond and replacing it with two C-O double bonds.
- The products are known as **carbonyl compounds**.
- As carvone is heated for a **prolonged time** with hot concentrated acidified potassium manganate (VII) **complete oxidation** will take place wherever possible.
- Compound **X** contains 3 carbonyl groups (three ketones).
- 2,4-dinitrophenylhydrazine (**2,4-DNPH**) reacts with carbonyl groups (ketones and aldehydes) and will therefore react with the **3 ketones**.

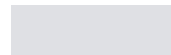


- The **CO₂** is formed from the oxidation of **-C=CH₂** to the **aldehyde** methanal followed by its oxidation to the **carboxylic acid** methanoic acid which in turn gets oxidised to **carbon dioxide**.

8

The correct answer is **A** because:

- **Trio-iodomethane** (also known as **iodoform**) forms a **yellow precipitate** with



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 - An alkaline solution of iodine is warmed with the substance to be tested.
 - The carbonyl compound is halogenated and the three hydrogen atoms in the CH_3 methyl group are replaced by iodine atoms.
 - This is a nucleophilic substitution reaction as the **iodine** acts as a **nucleophile** replacing the hydrogen atoms in CH_3 .
 - An intermediate is **hydrolysed** to form the yellow precipitate of tri-iodomethane.
- The **black** iodine solution will turn to **yellow precipitate** as crystals of trio-iodomethane are produced.
- Both ethanal and propanone contain a $\text{CH}_3\text{CO}-$ group and will therefore change the black iodine solution yellow.

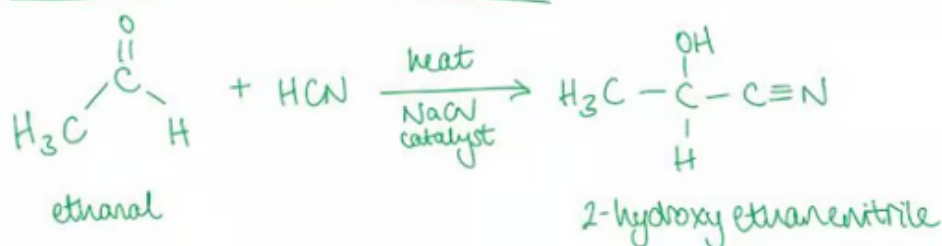
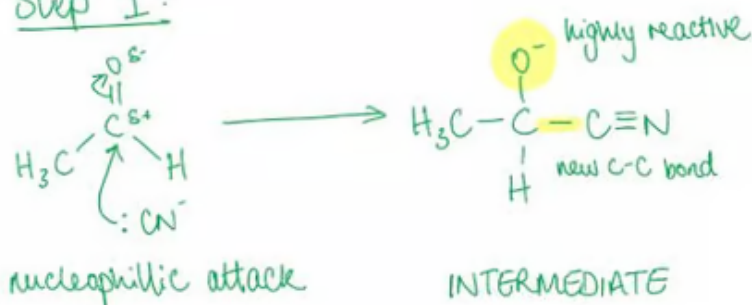
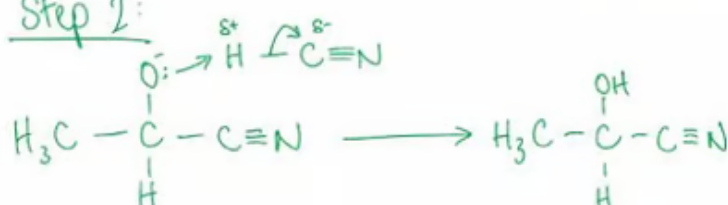
B, C & D are incorrect as **acidified potassium dichromate (VI)** is an **orange** oxidising agent. The reaction mixture turns **green** as the **orange $\text{Cr}_2\text{O}_7^{2-}$** ions (dichromate) are reduced to **green Cr^{3+}** ions when only **primary alcohols, secondary alcohols** and **aldehydes** can be oxidised by acidified potassium dichromate (VI). Propanone (a **ketone**) will therefore not turn the solution green.

C & D are incorrect as **Fehling's Solution** (CuSO_4) is a **blue** oxidising agent that changes when the blue solution of copper ions (Cu^{2+}) is reduced to **Cu_2O (s)** which is **red**. Only **aldehydes** can be oxidised by Fehling's Solution and propanone (a **ketone**) will therefore not turn the solution red.

9

The correct answer is **D** because:

- **Hydroxynitriles** can be formed from the **nucleophilic addition** reaction of **aldehydes** and **ketones** with **HCN, KCN** as catalyst and heat.
- **Ethanal** is an **aldehyde** and will react with hydrogen cyanide (the nucleophile) in the presence of sodium cyanide (as a **catalyst**) to form a hydroxynitrile.

Overall reaction mechanism:Step 1:Step 2:

↳ The proton can also be abstracted from water or dilute acid.

- The reaction mechanism shows that the **highly reactive intermediate** involves a **negatively charged oxygen** atom which quickly reacts with an aqueous H^+ ion to form an hydrogen- oxygen bond.
- The reaction mechanism shows that a new carbon-carbon bond is formed between, what used to be, the carbonyl-carbon and the carbon in the cyanide group.

A, B & C are incorrect as all statements are true.

10

The correct answer is **C** because:

- The presence of a **carbonyl group** in an **aldehyde** or **ketone** can be tested by adding a solution of 2,4-dinitrophenylhydrazine (**2,4-DNPH**).

- If an aldehyde or ketone is present, a **deep-orange precipitate** is formed.
 - The precipitate can be purified by **recrystallisation** and its melting point can be measured.
 - The **identity** of the precipitated compound can then be found by referring to the **melting point data** from which the specific aldehyde or ketone can be identified.
 - This reaction is an example of a **condensation reaction** in which two molecules join together and **eliminate a small molecule**.
- 